RESEARCH SUMMARY

X-only finite-grid DLV martingale simulation

WEEKLY RE-GRIDDING, EXACT TOWERS, AND MONTE CARLO EVIDENCE

A density-first risk-neutral law for the modeled finite option surfac

Y ≡ 1 • all 891 certified states • four 1W steps • 64-bit floats • antithetic paths

BOTTOM LINE

The finite-grid density tower is an exact martingale construction.

Re-gridding is an exact coordinate change for surviving fixed-cash contracts.

The unresolved modeling issue is absorbing tail support—not learned drift.

High-power result

A 100,000-path X-only four-week audit found covariance-aware p-values of 0.139, 0.406, 0.439, and 0.357 for the surviving original fixed-cash book plus spot. No original-contract horizon rejected zero drift at 5%.

Exact numerical identities

- All-state option-tower residual: 1.67e-16; density-tower residual: 6.49e-15.
- Direct cash-grid and re-gridded survivor prices agree pathwise to at most 2.22e-16.
- Four weekly rolls remain stable; boundary incidence reaches only 0.006% at week 4.

Scope: pure discrete DLV factor X. The independent Black factor is disabled exactly at the pricing and path layer.

Indices and probability law

The branch marginal and the pathwise conditional density are different objects.

m expiry	**i** front node	**j** later node	**b** branch
m = 1,...,M; T_m = t + m h	observed at T_1; S_(t+h) = s x_i	possible node at surviving T_m	one of 2^F surface-shock signs

State and branch symbols

- All formulas condition on one fixed current state; its state/batch index is suppressed. P_t and E_t mean probability and expectation given that state's current information.
- t is current calendar time; h is one week; M = 4. S_t = s is the X-only spot. Y ≡ 1, so there is no independent Black spot factor.
- There are N_x ordered support multipliers x_j. Node j represents the cash value s x_j. The indices i and j range from 0 to N_x−1 but play the distinct roles shown above.
- q_m(j) is the time-t risk-neutral probability mass at node j for expiry T_m. It has unit mass and unit mean: Σ_j q_m(j)=1 and Σ_j q_m(j)x_j=1.
- DLV means discrete local volatility; DLVsq is its square. F = 4 is the number of DLVsq shock factors; ε_b is branch b's vector of ±1 Rademacher signs; Γ_m maps those signs into expiry-m probability-mass shocks.
- π_b is the probability of branch b—not the circle constant. All 2^F = 16 branches are equally weighted here, so π_b = 2^(−F) = 1/16 and Σ_b π_b = 1.
- q_m^b(j) is the branch-conditioned marginal before front node i is observed. r_ij^(b,m) is the probability of later node j conditional on both branch b and observed front node i; that row is the actual pathwise survivor density.

$$P_t(B = b, I = i) = \pi_b q_1(i), \qquad q_1^b = q_1 \ (\Gamma_1 = 0)$$

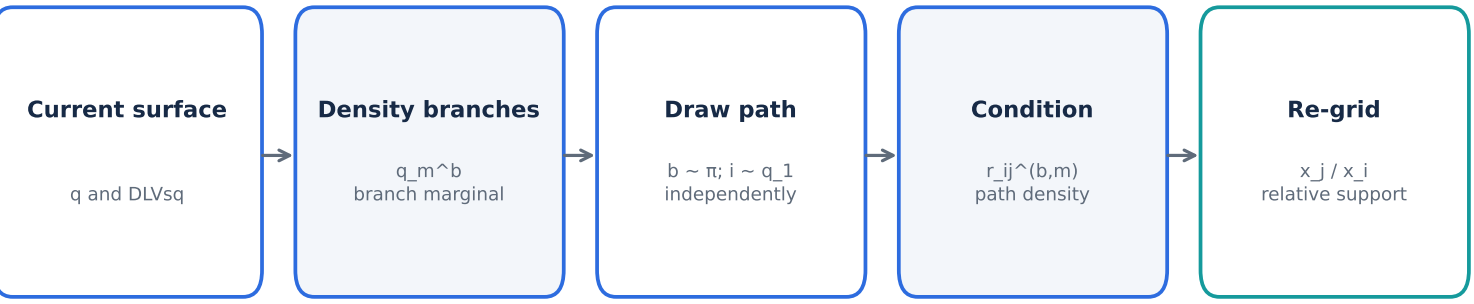
The implementation draws the branch B and front-node index I independently given current information. Conditioning and re-gridding then make the rolled surface depend on the realized spot. This exact law does not model contemporaneous correlation between the raw branch shock and the front-node draw.

METHOD

One-week construction

The method steps in density space, where the martingale restriction is linear, and only then recovers DLVsq coordinates. This replaces an approximate Euler drift calculation with an exact finite-step density mean.

$$q_m^b(j) = q_m(j) + \sqrt{h}\,[\Gamma_m \varepsilon_b]_j, \qquad \sum_b \pi_b q_m^b(j) = q_m(j)$$



Implementation sequence

- Derive density diffusion from the fixed-contract DLVsq loadings and the discrete SANOS operator.
- Enumerate symmetric Rademacher branches; reject a step if a branch is materially negative.
- Recover branch DLVsq algebraically from adjacent-expiry call values, then verify the decoder reproduces the branch density.
- Draw branch b with probability π_b and, independently, front node i with probability q_1(i); propagate the point mass at i to obtain r_ij^(b,m).
- Re-center support around realized spot. Old 2W–4W expiries become floating 1W–3W; a separate AR boundary supplies the newly issued 4W DLVsq.

What is fitted?

Only the new-4W boundary law is fitted: 25 principal-component factors and a weekly vector autoregression of order one, trained on 703 pairs and checked on 155 untouched 2025 pairs. The martingale transition itself is analytical.

PROOF

Exact re-grid martingale identity

For survivor expiry m ≥ 2, r_ij^(b,m) is the later-node probability after branch b and front node i are both known. It obeys a conditional martingale identity, an i-to-branch tower, and a branch-to-current tower.

$$\sum_j r_{ij}^{b,m} x_j = x_i, \quad \sum_i q_1(i) r_{ij}^{b,m} = q_m^b(j), \quad \sum_b \pi_b q_m^b(j) = q_m(j)$$

Here S_t=s, K is a fixed cash strike, and (a)^+=max(a,0). Re-gridding divides every inherited node by the same realized front multiplier.

$$S_{t+h} = s x_i, \qquad x_j'^{(i)} = x_j / x_i, \qquad S_{t+h} x_j'^{(i)} = s x_j$$

The conditional call value is therefore identical on the cash grid and the re-gridded relative grid.

$$C_{t+h}^{b,i,m}(K) = \sum_j r_{ij}^{b,m} (s x_j - K)^+$$

E_t means risk-neutral expectation conditional on current information. Applying both towers returns the current expiry-m call price.

$$E_t[C_{t+h}^m(K)] = \sum_b \pi_b \sum_i q_1(i) C_{t+h}^{b,i,m}(K) = \sum_j q_m(j)(s x_j - K)^+ = C_t^m(K)$$

Consequences

- The result holds for every strike and payoff represented on the finite support; puts follow by parity.
- No survivor-density interpolation is used. Re-gridding cannot introduce drift because it changes coordinates, not cash payoffs.
- This audit is pure X: the independent Black factor is identically one, so spot moves only when the front DLV node changes.

How zero drift is tested

Index a=1,...,n labels independent antithetic pairs; k=1,...,d labels spot/option return coordinates. R_ak is pair a's mean return in coordinate k, R-bar is the d-vector of sample means, and sigma-hat_k is the sample standard deviation using denominator n−1.

$$z_k = \sqrt{n}\,\bar{R}_k/\hat{\sigma}_k, \qquad J = n\,\bar{R}^T\hat{\Sigma}^+\bar{R}$$

Sigma-hat is the d × d sample covariance of pair observations. The superscript + denotes its truncated Moore–Penrose inverse: correlation eigenmodes at or below 0.01 are discarded. If ν modes remain, the reference is J approximately chi-square(ν); p is P(chi-square(ν) ≥ observed J) under zero population means.

SMALL p

Evidence against zero drift

Reject per test if p < 0.05

LARGE p

Compatible with MC noise

Not proof of exact equivalence

Decision thresholds

- One test: 0.05. Four pre-specified weekly looks: Bonferroni 0.0125.
- Worst |z| above two is not an automatic rejection across many correlated options.

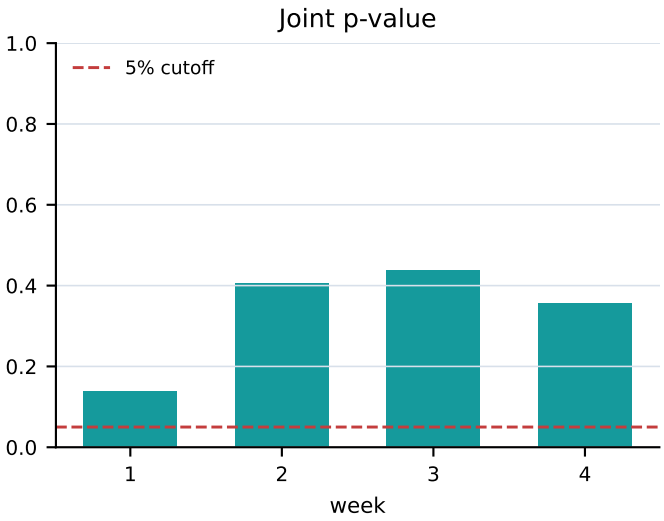
Calibration caveat

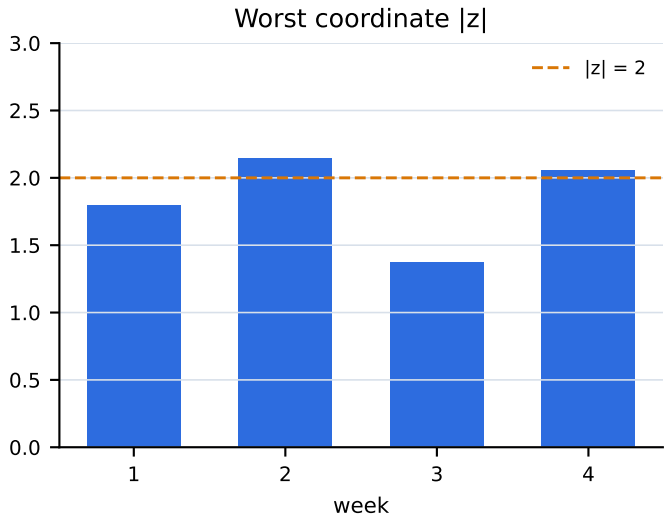
The option returns are highly collinear, and the eigenmodes are selected from a sample covariance. The nominal chi-square p-value can therefore move with sample size and cutoff. Exact tower identities and matched direct/re-grid residuals carry more weight than an isolated low p-value.

Surviving original contracts

X-only • 100,000 paths • 50,000 independent antithetic pairs • all 891 states • bp = 10^{-4}

Week	worst |z|	max mean (bp spot)	J	df	p
1	1.796	0.497	10.980	7	0.139
2	2.143	0.863	7.228	7	0.406
3	1.371	0.696	5.858	6	0.439
4	2.057	1.245	5.506	5	0.357





Interpretation

- All four original-contract covariance-aware tests fail to reject at 5%; each is also clear of the 0.0125 four-look cutoff.
- The maximum option mean is at most 1.245 basis points of initial spot.
- Exact residuals are 1e-13 or smaller, separating numerical consistency from Monte Carlo fluctuation.

Fixed-support / floating-coordinate decomposition

Matched X-only 100,000-path transition decomposition

Week	fixed-expiry p	direct = re-grid first-member p	full pair p	max price diff
1	0.366	0.684	0.139	0.0
2	0.538	0.163	0.128	2.22e-16
3	0.911	0.414	0.0208	2.22e-16
4	0.945	0.0808	0.546	2.22e-16

Direct inherited-grid and floating re-grid cash prices have identical z/J/p statistics to numerical precision. The pair column uses antithetic means; component columns use one independent member per pair.

Objectives and findings

- Fixed-grid objective: hold cash strikes and expiries fixed during each step and enforce the DLV surface-density mean exactly.
- Floating-grid objective: condition on the realized X node, divide support by that node, and repeat with no change to survivor cash values.
- Fixed-expiry branch p-values are 0.366, 0.538, 0.911, 0.945; its maximum option means are at most 0.0056 bp of spot.
- Rolling-book pair p-values are 0.139, 0.128, 0.0208, 0.546. Week 3 is below 0.05 but above the four-look cutoff 0.0125.
- The week-3 result cannot be re-grid drift: direct and re-grid returns are identical; p rises to 0.0879/0.295 at cutoffs 0.02/0.05, and split-covariance p is 0.214/0.338.

RESULT

This is one matched process—not two independently specified models.

Exact direct/re-grid equality rules out coordinate drift; the low p is covariance-mode sensitive.

Boundary behavior and assessment

At an extreme finite support node, a bounded martingale must stay there. If x_0 is the lowest node and the front draw selects it, every x_j is at least x_0.

$$\sum_j r_{0j}^{b,m} x_j = x_0 \quad \Longrightarrow \quad r_{00}^{b,m} = 1$$

The conditional density becomes a point mass and survivor options become intrinsic. Interior DLVsq coordinates are then price-unidentified: the positive-density formula divides by q and ceases to apply. The implementation's fallback continuation is a numerical convention invisible to the absorbed X law.

Observed in the X-only four-week run

- Boundary-node incidence: 0.003%, 0.003%, 0.003%, and 0.006% over weeks 1–4.
- Median effective front nodes: 34.81 at week 1 and 38.75 at week 4.
- At week 4, the first-percentile effective-node count is 7.44 and the 99th-percentile largest atom is 27.45%.

What would remove the artifact?

- A genuine unbounded tail continuation or dynamic support expansion before the boundary is selected.
- A tail atom with an internal conditional distribution, rather than treating it as one terminal point.
- Re-gridding alone cannot create missing downside or upside support while preserving the complete old option surface exactly.

Assessment

The approach establishes an internally consistent X-only risk-neutral law for the modeled finite surface. It preserves survivor prices exactly and remains stable through four weekly rolls. It is not a uniqueness result or a continuous-strike, unbounded-support construction. The next research step is explicit tail/support modeling.

Reviewed independently with QuantAgent: tower proof, re-grid identity, boundary degeneracy, and interpretation of z/J/p.